



**Q.** I have a JTAGICE MK-II programmer with me. How do I burn the bootloader in ATmega8 using it?

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Through E-mail

**A.** There are quite a few processes available for burning a bootloader in the required chip. The simplest is to directly use AtmelStudio and burn the .hex file of bootloader in the chip using an ISP programmer. The programmer you have comes with a JTAG-to-ISP header. Use this header to convert it to the ISP programmer and follow the

steps below:

1. Install latest AtmelStudio in your computer downloading from <http://www.atmel.in/tools/atmelstudio.aspx>.

2. Use an old Arduino Uno board and save yourself from making all the power and ISP connections on a bread board. Put the chip which needs the bootloader into the Arduino Uno board and connect the programmer to the ICSP header on the board.

3. Power up the board and the pro-

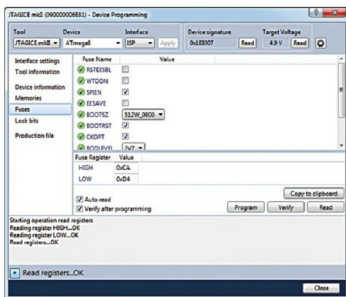


Fig. 3: Programming fuse bits

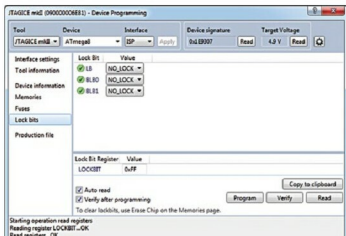


Fig. 4: Programming lock bits

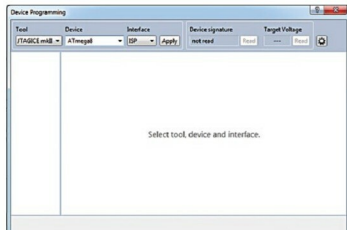


Fig. 1: Device programming window

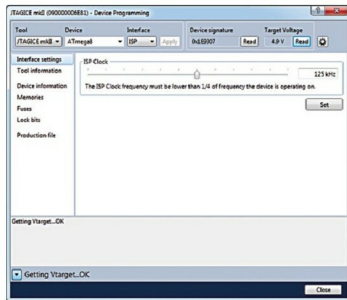


Fig. 2: Device signature read

grammer. You should see the green light on your programmer switched on.

4. Run Atmel Studio and click on Tools→Device programming. This will open device programming window as shown in Fig. 1.

5. Now press 'apply' after selecting appropriate values for tool, device and interface. Press 'read' and the window will show up various options and the device signature as shown in Fig. 2.

6. Now select the

ISP clock frequency as 125kHz if the chip is new and press 'Set.'

7. Click on memories option on the left and press erase button. After that, browse the .hex file of the bootloader under 'Flash' section. Press program button to burn the hex file in the chip. Once the programming is done, press verify button to verify the Flash.

8. Once the .hex file is programmed and verified, program the fuse bits as 0xA and 0xD4 as shown in Fig. 3.

9. Now the last step is to program the lock bits 0x0F as shown in Fig. 4 so that you do not wipe off the bootloader when you are flashing the firmware.

Answers compiled by EFY technical editor, Ankit Gupta. Letters and questions for publication may be addressed to Editor, Electronics For You, D-87/1, Okhla Industrial Area, Phase 1, New Delhi 110020 (E-mail: [editsec@efyindia.com](mailto:editsec@efyindia.com)) and should include name and address of the sender